

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.7: Surface Area and Volume of Solids — Teacher Reference (Pre-filled)

SUMMARY TABLE: CORE MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 2.7: Surface Area and Volume of Solids
Driving Question	DRIVING QUESTION: Why does the same amount of metal sheet make containers that hold very different amounts of grain — and how can a Kenyan artisan use mathematics to design the best solid shape for storage or construction?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Surface Area of Prisms: Same Metal Sheet, Different Grain Capacity?	Learners observed photographs and physical models of two metal storage containers from a posho mill in Eldoret — a cylindrical drum and a rectangular metal box — both made from the same total area of iron sheet. They also observed the constructed cardboard nets of rectangular and triangular prisms and measured the area of every individual face.	Learners learned that: (1) surface area is the total area of all faces of a three-dimensional solid, calculated by summing the areas of every face in the net; (2) the surface area formula for a rectangular prism is $SA = 2(lw + lh + wh)$; (3) the surface area formula for a triangular prism is $SA = 2(\frac{1}{2}bh_t) + (\text{perimeter of triangular face} \times \text{length of prism})$; (4) the rectangular metal box from the Eldoret posho mill ($l=120$ cm, $w=80$ cm, $h=100$ cm) has a surface area of $59,200$ cm ² , representing the total iron sheet used by the fundi.	Learners explained — partially — why a net represents the flat material used to construct a 3D container, and began to articulate that surface area (the iron sheet used) does not determine volume (how much grain fits inside). This explanation is incomplete: learners have posted the unresolved question 'Why does the cylinder hold more if the same sheet was used?' on the DQB, which drives the remainder of the sub-strand investigation.
Lesson 2 Surface Area of Pyramids: Designing Pyramid Roofs for Kenyan Structures	Learners constructed card nets of square-based and triangular pyramids, folded them into 3-D models, and measured both the slant height and the vertical height on their models. They observed that the slant height (the distance along the outer face from apex to base-edge midpoint) is always longer than the vertical height (the straight inner height), and that a square-based pyramid has one square face and four identical triangular faces.	Learners derived the surface area formula for a square-based pyramid: $SA = b^2 + 2bl$ (where b = base side length and l = slant height). They learned that slant height must be used — not vertical height — because it is the dimension that lies on the outer face of the pyramid, representing the material the metalsmith actually cuts. They applied Pythagoras' theorem to calculate slant height from given vertical height and base dimensions, and solved real-life problems involving the cost of iron sheet for pyramid-	Surface area measures the total area of the material covering the outside of a solid (like the iron sheet a metalsmith cuts to make a pyramid roof or a storage container). It does NOT measure how much space is inside the solid — that is volume. The posho mill owner's mystery cannot be solved by surface area alone: two containers can be made from the same area of iron sheet (same surface area) but hold very different amounts of maize (different volumes). To explain why the cylindrical drum holds more

		shaped roofs in Eldoret and Kericho.	grain than the rectangular box, we must investigate volume — which is what the next lessons will explore.
Lesson 3 Surface Area of Cones and Frustums: Unrolling the Cone to Find the Metal Sheet	Learners physically unrolled a paper cone into a sector and measured its arc length and the base circumference, observing that the two lengths are equal. They also watched a video of a Turkana metalsmith marking out a conical roof sector on a flat iron sheet.	The net of a cone is a sector of a circle with radius equal to the cone's slant height l . Because the arc of the sector equals the base circumference ($2\pi r$), the curved surface area of a cone is $CSA = \pi rl$. The slant height is found using Pythagoras: $l = \sqrt{r^2 + h^2}$. A frustum's curved surface area is $CSA = \pi(R + r)l$, derived by subtracting two conical surfaces. These formulas allow a metalsmith to calculate exactly how much iron sheet to cut before shaping a cone or frustum.	The posho mill artisan in Eldoret uses $CSA = \pi rl$ to determine the sheet metal needed for a conical funnel top, and $\pi(R + r)l$ for a frustum-shaped water-tank roof. This confirms that the same total area of metal sheet can be formed into very different shapes — and that the shape of a container, not just the sheet area used, determines how much grain it can hold. The driving question remains partially open: we have now compared surface areas of cones and frustums, but we have not yet compared their volumes to those of the cylinder and cuboid from Lessons 1–2. Volume calculations in Lessons 4–5 will complete the explanation of why the cylindrical drum holds more maize than the rectangular box.
Lesson 4 Surface Area of Spheres and Composite Solids	Learners wrapped a football with paper circles of matching radius and observed that exactly four circles were needed to cover the sphere's surface. They also observed labelled diagrams and worked through calculations for composite solids: a Nakuru grain silo (cylinder + cone) and a Mombasa water tank (cylinder + hemisphere), identifying which faces are exposed and which are hidden.	The surface area of a sphere is $SA = 4\pi r^2$ — equivalent to four flat circles of the same radius. To find the surface area of a composite solid, learners decompose it into known component shapes, identify all external (exposed) faces, exclude hidden internal faces, calculate each component area using the appropriate formula, and sum the results.	Learners explained that a metalsmith in Eldoret fabricating a spherical or dome-topped container can use $SA = 4\pi r^2$ and the decomposition algorithm to calculate precisely how much iron sheet is required. They connected this to the driving question by noting that different container shapes — even when built from the same area of iron sheet — enclose different amounts of space, and that knowing surface area formulae for all solid types is a prerequisite for comparing container efficiency. The DQB was updated with new questions about the relationship between surface area and volume, setting up Lessons 5 and 6.
Lesson 5 Volume of Prisms and Pyramids: How Much Grain Can the Box Really Hold?	Learners filled rectangular prism, triangular prism, and square-based pyramid models with sand and measured volumes using graduated containers, then posted group ratio values (pyramid volume $\div$ prism volume) on a class data board, observing that all ratios clustered near 0.33.	Learners derived that the volume of any prism equals base area multiplied by height ($V = B \times h$), and that the volume of a pyramid equals one third of the volume of a prism with the same base and height ($V = \frac{1}{3} \times B \times h$). They computed the volume of the posho mill owner's rectangular metal box as 1.44 m^3 and predicted that the cylindrical drum holds more grain despite being made	Equal surface area does not guarantee equal volume — the shape of the solid determines how much space is enclosed. The rectangular box encloses 1.44 m^3 of grain, but the cylindrical drum (with the same surface area of iron sheet) encloses a larger volume because the circular cross-section is more space-efficient than the rectangular cross-section for a given

		from the same area of iron sheet.	perimeter, a relationship that will be fully quantified in Lesson 6 when learners derive the volume formula for cylinders.
Lesson 6 Volume of Cones, Frustums and Spheres — Solving the Posho Mill Mystery	Learners observed a physical water-pouring demonstration in which a cone and a cylinder of identical base radius and height were used: it took exactly three full cones of water to completely fill the cylinder, establishing the 1:3 volume ratio empirically. They also watched a visual simulation of Cavalieri's principle showing equal cross-sectional areas between a hemisphere and a cylinder at every height.	The volume of a cone is $V = \frac{1}{3}\pi r^2 h$ — one third of the volume of a cylinder with the same base and height. The volume of a sphere is $V = \frac{4}{3}\pi r^3$, derived using Cavalieri's cross-sectional argument. The volume of a frustum is $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$. Using these formulas, learners back-calculated the cylindrical drum's radius from its known surface area of 59,200 cm ² , found $r \approx 38.7$ cm, and computed the drum's volume as approximately 1,483,000 cm ³ — compared to the rectangular box's volume of 1,440,000 cm ³ .	The posho mill mystery is now formally resolved: the cylindrical drum holds approximately 43,000 cm ³ — about 43 litres — more maize grain than the rectangular metal box, even though both were made from exactly the same area of iron sheet. This is because a circular cross-section encloses a greater area for a given perimeter than a rectangular cross-section does. Equal surface area does not guarantee equal volume — the geometry of the shape determines how efficiently a given amount of material encloses space. This is a principle that Kenyan artisans, grain store designers, and engineers can use to choose the most efficient container shape for a given amount of construction material.
Lesson 7 Volume and Surface Area of Composite Solids & Real-Life Applications	Learners observed the step-by-step decomposition of two composite solids — a cylinder topped with a hemisphere (Mwangi wa Kamau Water Tank, Makueni County) and a cylinder topped with a cone (Nakuru grain silo) — and completed structured observation tables identifying which surfaces are exposed (included in surface area) and which are internal (excluded), as well as calculating total volumes by summing component volumes.	Learners learned that: (1) a composite solid is decomposed into component solids for calculation; (2) the total surface area of a composite solid is the sum of only the EXPOSED surfaces — internal surfaces where two components are fused are excluded; (3) the total volume of a composite solid is the sum of the volumes of all component solids; (4) the same total surface area (same amount of metal sheet) can yield very different total volumes depending on which composite shape is chosen — directly explaining the posho mill owner's puzzle at a more advanced level.	Learners explained their understanding by: (1) solving the five-part Rift Valley Grain Silo Challenge (Mwalimu Chebet's silo in Eldoret) in groups, including cost calculations and a comparative volume analysis; (2) updating the Driving Question Board with explicit connections between composite solid calculations and the original phenomenon; (3) designing their own composite storage container with a fixed surface area constraint (≈ 4.8 m ²) and annotating their cumulative model diagram with the resulting volume, comparing it to the posho mill owner's cylindrical drum and rectangular box.
Lesson 8 Project Presentations, Final Model Building & Unit Consolidation	Learners observed each other's community container design projects displayed in a gallery walk around the classroom. They recorded surface area and volume values for cylindrical grain silos, rectangular water tanks, conical cooking-pot lids, and composite storage structures designed for Kenyan smallholder farming communities. They noted that different groups, working	Learners synthesised the full unit's content into a final annotated model: total surface area and volume formulae for cylinders ($TSA = 2\pi r^2 + 2\pi rh$; $V = \pi r^2 h$), cuboids ($TSA = 2(lb + bh + lh)$; $V = lbh$), cones, pyramids, frustums, spheres, and composite solids. They established that surface area measures the amount of material enclosing a solid while volume	Using the class consensus statement constructed during the Explain phase and the completed Driving Question Board, learners produced a full written explanation resolving the anchoring phenomenon: the Eldoret posho mill owner's cylindrical drum holds more maize than the rectangular metal box because, for the same total surface area of iron sheet, the cylindrical

	with similar amounts of material (surface area), produced containers with noticeably different holding capacities (volume), recreating the core puzzle of the Eldoret posho mill owner's cylindrical drum versus rectangular box at the scale of their own designed artefacts.	measures the space enclosed, and that these are independent properties determined by both size and shape. They articulated the isoperimetric principle — among solids with equal surface area, the cylinder encloses greater volume than a rectangular prism — as the mathematical resolution to the driving question. They connected these ideas to Kenyan careers including jua kali metalsmiths in Kamukunji, NCWSC water engineers, KEBS standards officers, and agricultural silo designers in the Rift Valley.	shape encloses a greater volume. This is because the circle — the cross-section of the cylinder — is the most area-efficient closed curve for a given perimeter (isoperimetric principle), meaning the circular base of the cylinder packs more capacity per unit of bounding material than the rectangular base of the box. Mathematically, two containers with equal TSA but different shapes will have different volumes, and the cylinder consistently wins this comparison against a same-height rectangular prism. A Kenyan artisan who understands surface area and volume formulae can therefore calculate the optimal solid shape and dimensions to maximise storage capacity for a given budget of metal sheet — turning an intuitive craft skill into a precise, reproducible mathematical design process.
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ATTRIBUTION AND LICENSING

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